FORCES OF FRICTION

1. A force of 170 N is needed to keep a 530-N wooden box sliding on a wooden floor. What is the coefficient of kinetic friction?

2. A cart on wheels weighs 2400 N. The coefficient of rolling friction between the wheels and floor is 0.16. What force is needed to keep the cart rolling uniformly(380 N)?

3. A wooden crate weighs 780 lb. What force is needed to start the crate sliding on a wooden floor when the coefficient of static friction is 0.40?

4. A piano weighs 4700 N. What force is needed to start the piano rolling across the floor when the coefficient of static friction is 0.23(1100 N)?

5. A force of 850 N is needed to keep the piano in Problem 48 rolling uniformly. What is the coefficient of rolling friction?

6. A dog sled weighing 750 lb is pulled over level snow at a uniform speed by a dog team exerting a force of 60 lb. Find the coefficient of friction(0.080).

7. A horizontal conveyor belt system has a coefficient of kinetic friction of 0.65. The motor driving the system can deliver a maximum force of 2.5 10^6 N what maximum total weight can be placed on the conveyor system?

8. A snowmobile is pulling a large sled across a snow-covered field. The weight of the sled is 3560 N. If the coefficient of friction of the sled is 0.12, what is the pulling force supplied by the snowmobile?

9. An automobile weighs 12 000 N and has a coefficient of static friction of 0.13. What force is required to start the auto rolling(1600 N)?

10. A light truck weighs 14 000 N with a coefficient of static friction of 0.140. What force is required to start the truck rolling?

11. A stake truck weighs 20000 N with a coefficient of static friction of 0.130. What force is required to start the truck rolling(2600 N)?

12. An alloy block is placed on a smooth composite table. If a force of 14.0 N is required to keep the 40.0-N block moving at a constant velocity, what is the coefficient of kinetic friction for the table and block (0.350)?

13. If the coefficient of friction between the tires of a car and the road is 0.300, what is the stopping distance of a car traveling at 60.0 mph?

14. A 2.00-kg toy accelerates from rest to 3.00 m/s in 8.00 s on a rough surface of μ_k= 0.300. Find the applied force **F.**

15. Rubber tires and wet blacktop have a coefficient of kinetic friction of 0.500. A pickup truck with mass traveling 30.0 m/s skids to a stop. (a)What is the size and direction of the frictional force that the road exerts on the truck? (b) Find the acceleration of the truck. (c) How far would the truck travel before coming to rest((a) – 3680 N (the – sign indicates the frictional force acts in opposite direction to that of the truck) (b)-4.91 m/s^2 (c) 91.6 m)?

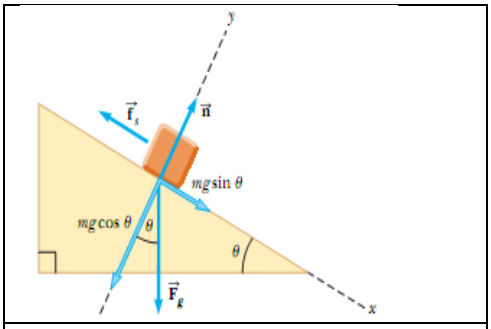
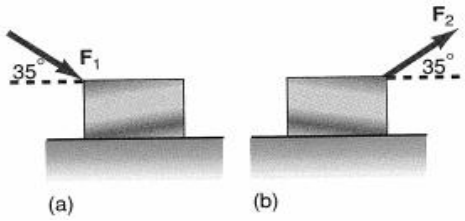


Figure 3

16. A 200-N container is to be pushed across a rough floor. The coefficient of static friction is 0.500 and the coefficient of kinetic friction is 0.400. What force is necessary to start the container moving, and what force is necessary to keep it moving at a constant velocity(100 N; 80 N)?

17. A 50.0-lb (1 lb = 4.45 N) box is to be moved along a rough floor at a constant velocity. The coefficient of friction is μ_K = 0.300. (a) What force F_1 must you exert if you push downward on the box as shown? (b) What force $\mathbf{F_2}$ must you exert if you pull upward on the box as shown? (c) Which is the better way to move the box(a. 23.2 lb or 103.24 N; b. 15.1 or 67.1 N); c. Method b is better. It requires less force to move the block)?

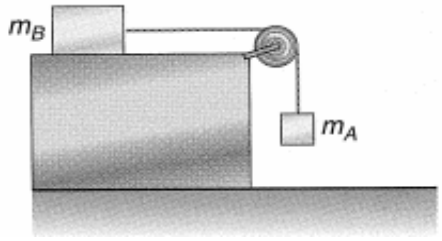


18. What force must act horizontally on a 20.0-kg mass moving at a constant speed of 4.00 m/s on a rough table of coefficient of kinetic friction of 0.300? If the force is removed, when will the body come to rest? Where will it come to rest?

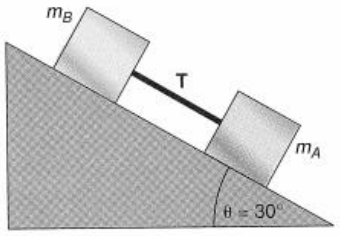
19. A mass of 2.00 kg is pushed up an inclined plane that makes an angle of 50.0° with the horizontal. If the coefficient of kinetic friction between the mass and the plane is 0.400, and a force of 50.0 N is applied parallel to the plane, what is (a) the acceleration of the mass and (b) its velocity after moving 3.00 m up the plane?

20. A boy coasts down a hill on a sled, reaching a level surface at the bottom with a speed of 7.0 m/s. If the coefficient of friction between the sled's runners and the snow is 0.050 and the boy and sled together weigh 600 N, how far does the sled travel on the level surface before coming to rest(50 m)?

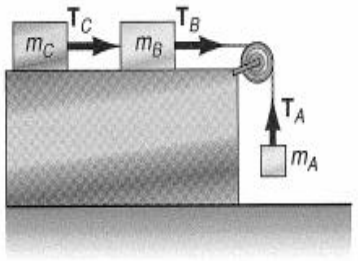
21. The two masses m_A = 20 kg and m_B = 20 kg are connected as shown on a rough table. If the coefficient of friction between block **B** and the table is 0.45, find (a) the acceleration of each block and (b) the tension in the connecting string.
(4.91 m/s^2; T=98.1 N)



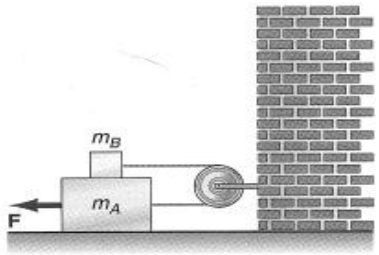
25. If m_A = 4.00 kg, m_B = 2.00 kg, μ_{kA} = 0.300, and μ_{kb} =0.400, find (a) the acceleration of the system down the plane and (b) the tension in the connecting string(2.07 m/s^2; 1.14 N).



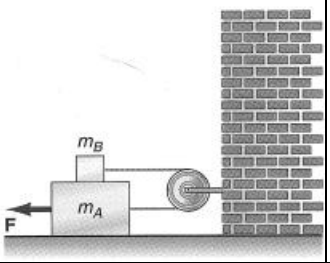
26. In the diagram $\boldsymbol{m_A}$ = 6.00 kg, $\boldsymbol{m_B}$ = 3.00 kg, m_c = 2.00 kg, $\boldsymbol{\mu_{kc}}$ = 0.400, and $\boldsymbol{\mu_{kB}}$ = 0.300. Find the magnitude of the acceleration of the system and the tension in each string(3.83 m/s^2; $T_A = T_B$= 35.8 N; T_C = 15.5 N).



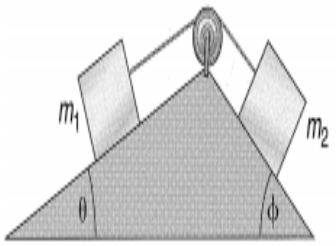
27. Find the force F that is necessary for the system shown to move at a acceleration of a = 0.5g m/s^2 if μ_k = 0.5 for all surfaces. The masses are m_A = 2 kg and m_B = 1 kg.
(F = 19.2 N)



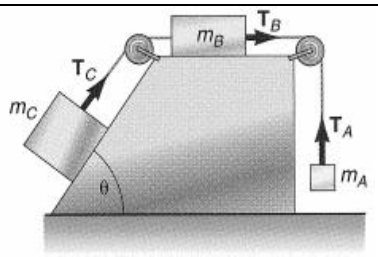
28. Find the force F that is necessary for the system shown to move at constant velocity if μ_k = 0.3for all surfaces. The masses are m_A = 2 kg and m_B = 1 kg. (F = 11.8 N)



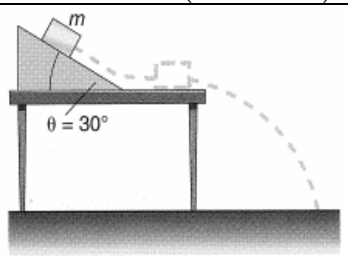
29. Two blocks m_1 = 20.0 kg and m_2 = 10.0 kg are connected as shown on a frictionless plane. The angle $\boldsymbol{\theta}$= 25.0° and $\boldsymbol{\phi}$ = 35.0°. Find the acceleration of each block and the tension in the connecting string. (a = 0.886 m/s^2; T = 65.1 N)



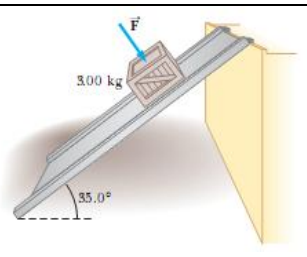
30. If $\boldsymbol{m_A}$ = 6.00 kg, $\boldsymbol{m_B}$ = 2.00 kg, $\boldsymbol{m_c}$ = 4.00 kg, θ = 58^0 and the coefficient of kinetic friction for the surfaces are $\boldsymbol{\mu_{kB}}$ = 0.300 and μ_{kc} = 0.200 find the magnitude of the acceleration of the system shown in the diagram and the tension in each string(1.25 m/s^2; $T_A = T_B$= 51.3 N; T_C = 42.9 N).



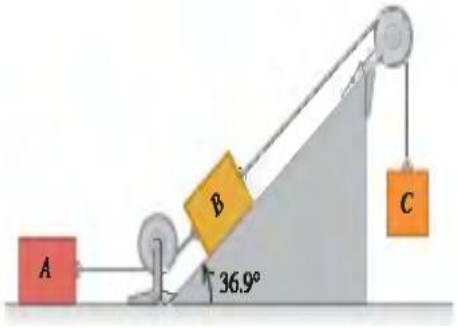
31. A block m = 0.500 kg slides down a frictionless inclined plane 2.00 m long. It then slides on a rough horizontal table surface of $\boldsymbol{\mu_K}$ = 0.300 for 0.500 m. It then leaves the top of the table, which is 1.00 m high. How far from the base of the table does the block land? (x = 1.34 m)



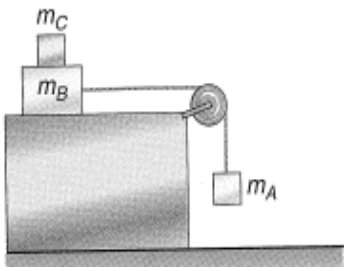
32. The coefficient of static friction between the 3.00-kg crate and the 35.0 incline of Figure is 0.300. What minimum force must be applied to the crate perpendicular to the incline to prevent the crate from sliding down the incline (32.1 N)?



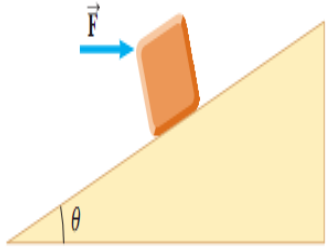
33. Blocks A, B, and C are placed as in Fig. and connected by ropes of negligible mass. Both A and B weigh 25.0 N each, and the coefficient of kinetic friction between each block and the surface is 0.35. Block C descends with constant velocity. (a) Draw two separate free-body diagrams showing the forces acting on A and on B. (b) Find the tension in the rope connecting blocks A and B. (c) What is the weight of block C? (d) If the rope connecting A and B were cut, what would be the acceleration of C((b) 8.75; (c) 30.8; (d) 1.54 m/s^2)?



34. In the diagram, block B rests on a frictionless surface but there is friction between blocks B and C. $m_A = 2$ kg, $m_B = 3$ kg, and $m_C = 1$ kg. Find (a) the magnitude of the acceleration of the system and (b) the minimum coefficient of friction between blocks C and B such that C will move with B((a) 3.27 m/s^2; (b) 0.333)



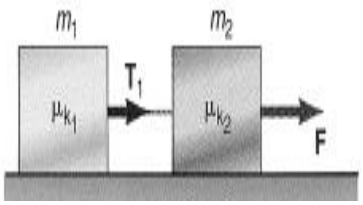
35. A 2.00-kg block is held in equilibrium on an incline of angle $\theta = 60.0^0$ by a horizontal force F applied in the direction shown in Figure. If the coefficient of static friction between block and incline is $\mu_s = 0.300$, determine (a) the minimum value of F and (b) the normal force exerted by the incline on the block.
((a) F_min = 10.5 N;
(b) N = 13.6 N)



36. The person in Figure weighs 170 lb. Each crutch makes an angle of 22^0 with the vertical (as seen from the front). Half of the person's weight is supported by the crutches, the other half by the vertical forces exerted by the ground on his feet. Assuming that he is at rest and that the force exerted by the ground on the crutches acts along the crutches, determine (a) the smallest possible coefficient of friction between crutches and ground and (b) the magnitude of the compression force supported by each crutch((a) 0.404; (b) 45.8 lb).



37. What force is necessary to pull the two masses at constant speed if $m_1 = 2$ kg, $m_2 = 5$ kg, $\mu_{k1} = 0.3$ and $\mu_{k2} = 0.2$? What is the tension T_1 in the connecting string? ($T_1 = 11.8$ N)



38. Find (a) the magnitude of the acceleration of the system shown if $\mu_{kA} = 0.3$, $\mu_{kB} = 0.2$, $m_A = 3$ kg, and $m_B = 5$ kg, (b) the velocity of block A at 0.5 s.
((a) a = 2.45 m/s²; (b) v_A = 1.23 m/s)

